

RICHARD MAGNOTTI

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Summary

Software Engineer | Python, SQL, REST APIs, AWS | Amazon Alexa Prize semifinalist (\$250K grant)

Education

Rutgers University – PhD in Computer Science (GPA 3.97/4.0) Aug. 2021 – Present

University of Rochester – MS in Computer Science (GPA 3.66/4.0) May 2021

Southern CT State University – BS in Physics, w/ Honors; Minors: Computer Science, Mathematics Dec. 2017

Technical Skills

Languages: Python, SQL, Java, JavaScript, Bash, HTML/CSS

Backend/Systems: REST APIs, Node.js, Express, Flask, PostgreSQL, WebSockets (Socket.IO), Docker (Compose), Linux, ETL, OOP/design patterns, Agile/Scrum

Cloud/Tools: Git, AWS (EC2, ECS, Lambda, S3, DynamoDB, IAM, CloudWatch), Selenium, SLURM, Trello, Confluence

AI Coding Tools: Claude Code, OpenAI Codex, Cursor, GitHub Copilot

Relevant Experience

Real-Time AI Backend for Live Training Application Sep. 2024 – Dec. 2025

Project Manager & Software Developer. New Brunswick, NJ

- Built a real-time, multi-user chat backend (**Python, Flask, Socket.IO/WebSockets**) for a teacher-training simulator with student personas driven by a self-hosted **Llama 3.1 8B (Hugging Face Transformers)**.
- Engineered a thread-safe pipeline so 10+ concurrent sessions share one GPU-resident **LLM**, using **dynamic batching** to merge in-flight requests into one forward pass, with correlation IDs matching results to callers.
- Implemented a **YAML-driven** finite-state-machine dialogue engine (2,600+ states across 5 personas) that routes free-text input to the correct narrative branch via **LLM classification**.
- Built a **CSV-to-YAML compiler** so a psychology research team could author new scenarios from exported flowcharts with no code changes, running recurring demos and live trials with these non-technical stakeholders.

Amazon Alexa Prize TaskBot Challenge 2 Jan. 2023 – July 2023

Technical Lead. New Brunswick, NJ

- Led 9-person team (ML, backend, frontend) across the full SDLC using **Agile/Scrum** to build a voice+visual cooking/DIY assistant on **AWS (EC2, ECS, Lambda, S3, DynamoDB, IAM, CloudWatch)**, reaching *semifinals*.
- Co-wrote competitive proposal, selected as 1 of 10 teams worldwide, securing \$250K in Amazon funding.
- Architected an event-driven dialogue backend: a policy layer routed intents by phase (search vs. task execution) to response generators (QA, step navigation, recommendations) that teammates iterated in parallel.
- Designed persistent state management on **DynamoDB**, working around the Alexa framework's stack-size limits to keep conversation history, on-screen selections, and task progress resumable across sessions.
- Mined **CloudWatch** logs from hundreds of rated live sessions to diagnose failure modes (intent mismatches, flow errors, ASR misroutes), prioritizing fixes by how often each issue hit real users.

Orolia (now Safran) June 2020 – Aug. 2020

Software Engineer Intern. Rochester, NY

- Prototyped the company's first ML-based GPS spoof/jam detection (PyTorch DNN over Kalman-filter innovation-residual features), built Bash pipeline that queried the hardware and aggregated its training data.

My Gene Counsel April 2018 – Aug. 2019

Software Automation Engineer Intern. New Haven, CT

- Built the startup's automated regression testing from scratch, **Selenium (Python)** suites covering patient and admin portals, on an engagement scoped for 6-8 weeks and extended to 16 months.

Personal Projects

WhoDis: Full-stack support-ticket system. Built a containerized **Node/Express REST API** backed by **Postgres** with a **Docker** Compose dev environment, handling tickets from browser submission through persistence and retrieval on a single-developer build. [github.com/R-Magnotti/WhoDis]